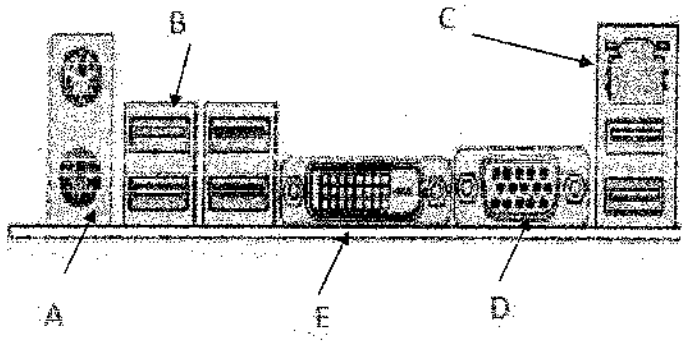
Higher National Diploma in Engineering
First Year, First Semester Examination – 2019
IT1101/BS11103/QS11053- Information Technology I

Instructions for Candidates:	No. of questions :06
Answer five questions only	No. of pages : 05
Each question carries 20 marks.	Time : Three (03) Hours

Q1.

I. Identify the following ports on the mother board. **(05 Marks)**



II. Name a device that can be connected on the above ports. **(05 Marks)**

III. List three (03) characteristics of microcomputers. **(03 Marks)**

IV. Write the meaning of the following abbreviations. **(04 Marks)**

a. CPU
b. UPS
c. SATA
d. USB

V. Compare the Dot Matrix printer and Laser printer on the followings. **(03 Marks)**

a. Cost
b. Speed
c. Quality of the printing

Q2.

An Engineer has decided to purchase a desk top computer to do his works efficiently. His main works are as follows:

- Prepare the Engineering Drawings
- Maintain Inventory
- Prepare Salary sheets
- Sending reports to the head of the department
- Prepare presentation about works done.

Write the answers for the following questions by considering the above scenarios.

I. What is the recommended type, the speed of processor and bus width for the above desktop computer? **(03 Marks)**

II. Give three (03) important input devices and explain their usage to fulfill above requirements. **(03 Marks)**

III. Give two (02) important output devices and explain their usage to fulfill above requirements. **(02 Marks)**

IV. What is the type of RAM and the suitable capacity? **(04 Marks)**

V. What is the suitable Operating System? Explain the reason. **(04 Marks)**

VI. Identify the suitable software for the following tasks. **(04 Marks)**

   a. Engineering Drawings
   b. Preparation of Salary sheets and maintain the inventory
   c. Prepare presentation.
   d. Sending reports to the head of the department

Q3.

I. Explain the Browser by giving two examples of them. **(04 Marks)**

II. Write the meaning of the following abbreviations. **(04 Marks)**

   a. URL
   b. HTML
   c. HTTP
   d. WWW

III. Consider the following URL to answer the questions below.

http://www.sliate.ac.lk/component/users/index.html?view=login

   a. What is the domain name? **(02 Marks)**
   b. What is the protocol used on the above URL? **(02 Marks)**
   c. What is the file name? **(02 Marks)**
   d. What is the parameter? **(02 Marks)**

IV. Suppose 10 computers are connected by using the followings topologies. Find the number of cables required for each. **(04 Marks)**

   a. BUS
   b. STAR
   c. RING
   d. MESH

Q4.

I. Write three (03) features of a spread sheet software. **(03 Marks)**

II. Give two examples of spread sheet software. **(02 Marks)**

III. Explain absolute cell reference and relative cell reference. **(03 Marks)**

IV. Ceylon Electricity Board (CEB) prepares the monthly bills for their customers by using Ms Excel sheet as follows.

E7 T02

	A	B	C	D	E	F	G	H	I	J	K	M
1												
2					Monthly Bill							
3					CEB							
4							Fixed Charge		Rs 540.00			
5	Customer				Transformar ID	Unit Consumed	Units From Solar system	Net Units	Charge for units consume	Outstanding	Total Amount	
6	ID	First Name	Last Name	Name with Initials								Transformar ID
7	1	Valli	Kumara		T02	150	55			Rs 120.00		T01
8	2	Santa	Malini		T01	213	0			Rs 0.00		T02
9												T03
10												
11												

CEB maintains the details of Customer ID, First Name, and Last Name. also for the usage they store the Transformer ID, Units Consumed, Units from Solar System, and Outstanding value. Charge for Units Consumed is calculated as follows:

Net Units	Charge per unit
1- 50	Rs 8.00
51-100	Rs. 9.00
Above 100	Rs 10.00

Write formula/function for the following using appropriate cell references.

a. What is the address of the current cell? **(01 Mark)**

b. What is the suitable cell format of the cell **I4** **(01 Mark)**

c. User select transformer ID from a drop down list. The items of the list is taken from the cell range **M7: M9.** Explain how to do it. **(02 Mark)**

d. Calculate the **Net Units in cell H7.**

Hint: Net Unit= Units Consumed - Unit from Solar system. **(02 Mark)**

e. Calculate **Charge for units Consumed in cell I7.** **(03 Marks)**

f. Calculate **Total Amount in cell K7.** **(03 Marks)**

Hint: Total Amount= Charge for units Consumed +Fixed Charge + Outstanding.

Q5.

I. List four (04) features of a DBMS software. **(02 Marks)**

II. Explain the following keys. **(02 Marks)**

a. Primary Key

b. Foreign Key

III. Answer the questions by considering the following MS Access database tables that are used to pay salary for the employees of a construction company.

Employee

Employee_Id	Employee_Name	Salary_Code
E01	Sunil	S01
E04	Kumar	S03

BasicSalary

Salary_Code	Job_Title	Basic_Salary
S01	Manager	120000.00
S02	Engineer	100000.00
S03	Technician	30000.00
S04	Consultant	115000.00
S05	Supervisor	80000.00

Allowance

Employee_Id	Festival	OT
E01	1000.00	2000.00
E04	800.00	2400.00

Deduction

Employee_Id	Loan_Payment
E01	0.00
E04	1000.00

a. Name a suitable data type for the following fields. **(04 Marks)**

- Employee_Id
- Employee_Name
- Salary_Code
- Basic_Salary

b. Identify the primary keys for the **Employee** and **BasicSalary** tables. **(02 Marks)**

c. Identify the foreign keys for the tables **Employee** and **BasicSalary**. **(01 Mark)**

d. Write down a SQL statement to add new employee to the Employee table (Employee name: Rani, Employee ID: E02, and Salary code: S02) **(02 Marks)**

e. Write down a SQL statement to display Employee ID, Employee name, and Job Title on alphabetic order of Employee name.

(03 marks)

f. Consider the following calculation:

Total Salary = Basic Salary + Festival +OT.
The Tax deduction is 5% of the Total salary.
Net Salary = Total Salary – Loan Payment – Tax.

Write down a SQL statement to display the followings: **(04 Marks)**

Employee Id, Employee Name, Basic Salary, Festival, OT, Total Salary, Loan Payment, Tax, and Net Salary on the order of Employee Id

Q6.

I. Write four (04) main functions of an Operating system. **(04 Marks)**

II. Describe the usage of File Explorer/Window Explorer. **(04 Marks)**

III. Compare MS Windows and Linux. **(04 Marks)**

IV. Give four (04) disadvantages of the Internet/Internet Applications. **(04 Marks)**

V. What are the short cut keys for the following commands in MS Word? **(04 Marks)**

a. Print
b. Paste
c. Undo
d. Increase the font size